

5. (a) A bow and arrow is a kind of projectile weapon. The string of a bow is drawn taut by a hunter with a force of 60 N and an arrow of mass 0.2 kg is held stationary as shown in Figure 5.1.

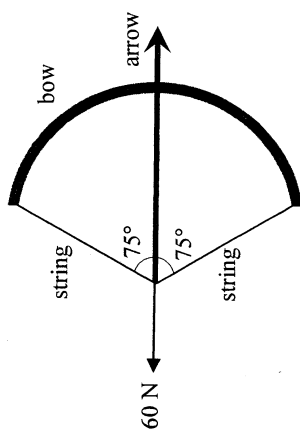


Figure 5.1

- (i) Find the tension of the string. Neglect the weight of the arrow. (2 marks)

- (ii) Estimate the energy stored in the taut string if the initial speed of the arrow is 45 m s^{-1} when released. Assume that the bow is rigid and neglect the mass of the string. (2 marks)

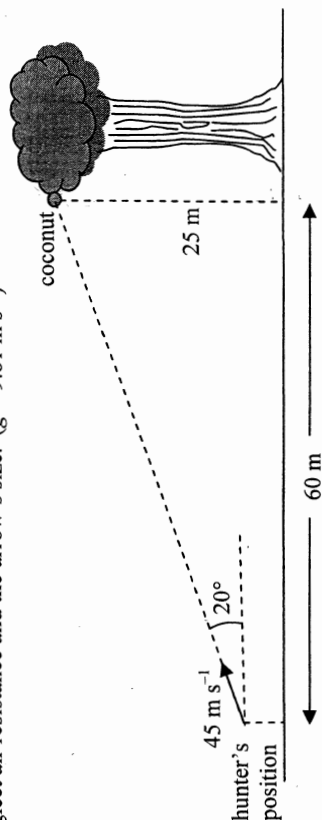
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- *(b) The hunter stands at about 60 m away from a tree as shown in Figure 5.2. He uses the bow to release the arrow in order to shoot a coconut held by a monkey (not shown in the figure) in the tree. The coconut is at a height of 25 m from the ground. The hunter aims directly at the coconut and the arrow leaves the bow at a speed of 45 m s^{-1} making an angle of 20° to the horizontal. At the moment the hunter releases the arrow, the monkey drops the coconut such that it falls vertically from rest. Neglect air resistance and the arrow's size. ($g = 9.81 \text{ m s}^{-2}$)

Figure 5.2



- (i) Find the time taken for the arrow to hit the coconut. (2 marks)

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- (ii) Find the height of the coconut from the ground at the moment the arrow hits it. (2 marks)

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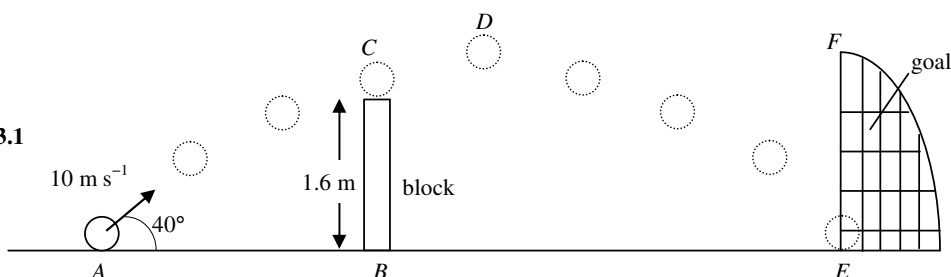
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3. A ball is kicked and moves with an initial velocity of 10 m s^{-1} at an angle of 40° to the horizontal. The ball then just passes a block of height 1.6 m , reaching the highest point D , and finally hits the ground at E as shown in Figure 3.1. (Neglect air resistance and the size of the ball.)

Figure 3.1



- (a) Draw an arrow to indicate the direction of acceleration of the ball at C . (1 mark)
- *(b) For a projectile of initial velocity u that makes an angle θ with the horizontal, show that its horizontal range is given by $\frac{u^2 \sin 2\theta}{g}$. Hence, or otherwise, find another angle of projection such that the ball can still reach E with the same initial speed of 10 m s^{-1} . (4 marks)
(Given: $2 \sin \theta \cos \theta = \sin 2\theta$)

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- (c) Calculate the speed of the ball at C . (2 marks)

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